

EXERCISE 8.2

Arithmetic Progressions

Questions with Step-by-Step Solutions

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Key Idea

If an AP has first term a and common difference d , then its n th term is

$$t_n = a + (n - 1)d$$

where

- a = first term,
- d = common difference,
- n = position of the term.

Question 1

For the AP

3, 8, 13, 18, ...

find the 10th and 26th terms.

Solution

Given AP:

$$3, 8, 13, 18, \dots$$

Therefore,

$$a = 3$$

and

$$\begin{aligned}d &= 8 - 3 \\&= 3 + 5 - 3 \\&= 5.\end{aligned}$$

We know,

$$t_n = a + (n - 1)d$$

For the 10th term:

$$\begin{aligned}t_{10} &= 3 + (10 - 1)(5) \\&= 3 + 45 \\&= \boxed{48}.\end{aligned}$$

For the 26th term:

$$\begin{aligned}t_{26} &= 3 + (26 - 1)(5) \\&= 3 + 125 \\&= \boxed{128}.\end{aligned}$$

Final Answer

$$\boxed{t_{10} = 48, \quad t_{26} = 128}$$

Question 2

For the AP

$$21, 18, 15, \dots$$

find which term is -81 .

Also determine whether 0 is a term of this AP.

Solution

Here,

$$a = 21$$

and

$$\begin{aligned}d &= 18 - 21 \\ &= -3.\end{aligned}$$

Using

$$t_n = a + (n - 1)d$$

we get

$$\begin{aligned}t_n &= 21 + (n - 1)(-3) \\ &= 21 - 3n + 3 \\ &= 24 - 3n.\end{aligned}$$

First, check for -81 :

$$\begin{aligned}24 - 3n &= -81 \\ -3n &= -105 \\ n &= 35.\end{aligned}$$

Thus, -81 is the 35th term.

Now check for 0:

$$\begin{aligned}24 - 3n &= 0 \\ 3n &= 24 \\ n &= 8.\end{aligned}$$

Since $n = 8$ is a positive integer, 0 is indeed a term.

Final Answer

-81 is the 35th term

0 is the 8th term

Question 3

For the AP

$$11, 8, 5, 2, \dots$$

find its n th term and write its recursive rule.

Solution

Here,

$$a = 11$$

and

$$\begin{aligned}d &= 8 - 11 \\ &= -3.\end{aligned}$$

Therefore,

$$\begin{aligned}t_n &= a + (n - 1)d \\ &= 11 + (n - 1)(-3) \\ &= 11 - 3n + 3 \\ &= \boxed{14 - 3n}.\end{aligned}$$

Recursive rule:

The first term is

$$\boxed{t_1 = 11}$$

Each next term is obtained by subtracting 3:

$$\boxed{t_n = t_{n-1} - 3, \quad n \geq 2}$$

Final Answer

$$\boxed{t_n = 14 - 3n}$$

$$\boxed{t_1 = 11, \quad t_n = t_{n-1} - 3}$$

Question 4

An AP has 50 terms. Its 3rd term is 12 and its last (50th) term is 106.
Find the 29th term.

Solution

Let the first term be a and common difference be d .

Using

$$t_n = a + (n - 1)d$$

For the 3rd term:

$$\begin{aligned}t_3 &= a + 2d \\a + 2d &= 12.\end{aligned}$$

For the 50th term:

$$\begin{aligned}t_{50} &= a + 49d \\a + 49d &= 106.\end{aligned}$$

Subtracting the two equations:

$$\begin{aligned}(a + 49d) - (a + 2d) &= 106 - 12 \\47d &= 94 \\d &= 2.\end{aligned}$$

Now,

$$\begin{aligned}a + 2d &= 12 \\a + 4 &= 12 \\a &= 8.\end{aligned}$$

Therefore,

$$\begin{aligned}t_{29} &= a + (29 - 1)d \\&= 8 + 28(2) \\&= 8 + 56 \\&= \boxed{64}.\end{aligned}$$

Final Answer

$$\boxed{t_{29} = 64}$$

Question 5

Consider all two-digit positive integers that are divisible by 3.

Find:

- (a) the number of such integers;
- (b) their total sum.

Solution

The two-digit multiples of 3 form the AP

$$12, 15, 18, \dots, 99$$

Therefore,

$$a = 12, \quad d = 3, \quad t_n = 99.$$

Using

$$t_n = a + (n - 1)d$$

we get

$$99 = 12 + (n - 1)3$$

$$87 = 3(n - 1)$$

$$29 = n - 1$$

$$n = 30.$$

Thus, there are

$$\boxed{30}$$

two-digit multiples of 3.

Now, sum of n terms of an AP is

$$S_n = \frac{n}{2}(a + l)$$

where l is the last term.

Therefore,

$$S_{30} = \frac{30}{2}(12 + 99)$$

$$= 15(111)$$

$$= \boxed{1665}.$$

Final Answer

$$\boxed{\text{Number of integers} = 30}$$

$$\boxed{\text{Sum} = 1665}$$

Question 6

A person's annual salary starts at 5,00,000 and increases by 20,000 every year. Determine the year in which the salary becomes 7,00,000.

Solution

The salaries form an AP:

$$5,00,000, 5,20,000, 5,40,000, \dots$$

Therefore,

$$a = 5,00,000$$

and

$$d = 20,000.$$

We need the term equal to

$$7,00,000.$$

Using

$$t_n = a + (n - 1)d$$

we get

$$7,00,000 = 5,00,000 + (n - 1)(20,000)$$

$$2,00,000 = (n - 1)(20,000)$$

$$10 = n - 1$$

$$n = 11.$$

Therefore, the salary becomes 7,00,000 in the

11th year

Final Answer

Salary reaches 7,00,000 in the 11th year.

Question 7

Suppose the number of objects arranged in successive rows is

$$1, 2, 3, 4, \dots, 25.$$

Find the total number of objects in all 25 rows.

Solution

We need to find

$$1 + 2 + 3 + \dots + 25.$$

The sum of the first n natural numbers is

$$S_n = \frac{n(n+1)}{2}$$

Here,

$$n = 25.$$

Therefore,

$$\begin{aligned} S_{25} &= \frac{25(25+1)}{2} \\ &= \frac{25 \times 26}{2} \\ &= 25 \times 13 \\ &= \boxed{325}. \end{aligned}$$

Final Answer

$$\boxed{325}$$

Key Idea**Quick Revision**

$$t_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}(a + l)$$

$$1 + 2 + 3 + \cdots + n = \frac{n(n + 1)}{2}$$

Remember:

AP $\rightarrow$ Same Difference

Find d first, then use the formula!

“Find the first term, find the difference, and the AP becomes easy!”

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